

EXPERIMENT - 3

Title: 3D Modelling, Transformations (Scaling, Rotation, Translation) using AutoCAD

1. Objective

To create 3D models of basic objects and perform transformations such as scaling, rotation, and translation using AutoCAD.

2. Software Required

AutoCAD (3D workspace)

3. Theory (Short)

3D modeling is the process of creating a mathematical representation of a 3D object.

Transformations include:

Translation → Moving object

Rotation → Rotating object

Scaling → Changing size

4. Initial Setup in AutoCAD

Step 1: Open 3D Workspace

Open AutoCAD

Switch workspace:

Click Workspace → Select 3D Modeling

Step 2: Set Units

Command: UNITS

Set:

Type: Decimal

Scale: Millimeters

Step 3: Set View

Use VIEWCUBE or:

Command: V → Select SW Isometric

5. Basic 3D Commands

Command	Function
BOX	Create rectangular solid
CYLINDER	Create cylinder
CONE	Create cone
SPHERE	Create sphere
EXTRUDE	Convert 2D → 3D
UNION	Join objects
SUBTRACT	Remove material
INTERSECT	Common portion

6. PROCEDURE

6.1 Creating Basic 3D Models

A. Create a Box

Steps:

Type BOX → Enter

Specify base corner → 0,0,0

Specify length → 80

Specify width → 50

Specify height → 40

B. Create a Cylinder

Steps:

Type CYLINDER → Enter

Specify center → 100,0,0

Radius → 20

Height → 60

C. Create a Cone

Steps:

Type CONE → Enter

Specify base center

Radius → 20

Height → 60

6.2 3D Transformations

A. Translation (Move Object)

Command: MOVE

Steps:

Select object

Press Enter

Specify base point

Specify new location

👉 Example: Move object from (0,0,0) to (50,50,0)

B. Rotation

Command: ROTATE

Steps:

Select object

Specify base point

Enter angle (e.g., 45°)

👉 Object rotates about axis

C. Scaling

Command: SCALE

Steps:

Select object

Specify base point

Enter scale factor

1 → Increase size

<1 → Decrease size

👉 Example: Scale = 2 (double size)

6.3 Advanced 3D Operations

A. Extrude

Steps:

Draw 2D shape (circle/rectangle)

Type EXTRUDE

Select object

Enter height

B. Boolean Operations

UNION

Combines objects into one

SUBTRACT

Removes one object from another

INTERSECT

Keeps common volume

7. Visualization

Step:

Command: SHADEMODE or VISUALSTYLES

Select:

Conceptual

Realistic

8. UCS (User Coordinate System)

Command: UCS

Use to:

Change drawing plane

Align objects in 3D

9. View Control

Command Use

ORBIT Rotate view

PAN Move view

ZOOM Zoom in/out

10. Precautions

Always work in correct UCS

Use proper coordinates

Check object orientation

Avoid overlapping solids

11. Result

3D models were successfully created and transformed using AutoCAD tools like MOVE, ROTATE, and SCALE.

12. Viva Questions

What is 3D modeling?

Difference between 2D and 3D drawing

What is extrusion?

Define scaling, rotation, translation

What is UCS?

What is Boolean operation?